Integrating land-use and spatial factors into energy modeling: Onshore wind and agriculture co-location in Prince Edward Island

Narges Sefid, Taco Niet

School of Sustainable Energy Engineering, Simon Fraser University, Surrey, British Columbia

Introduction

- This work examines the synergies of electricity generation costs, emissions, and land use under the co-location of onshore wind and agriculture.
- We studied Prince Edward Island province with the TEMOA-ACES (Atlantic Canada Energy System model) [1][2].
- Scenarios investigated in this work are the co-location, where onshore wind is located on cropland, and the base scenario, which does not include co-location. A range of moderate to extreme land constraints is then applied to both.
- Results and findings provide insights for policymakers in Atlantic Canada.

Literature Gaps Addressed

- Renewables apply a major land footprint to the energy system due to their high land intensity and extensive development [3].
- Onshore wind has the highest land intensity among other energy production technologies if we consider the spacing between turbines [4].
- Cropland offers potential co-location opportunities with onshore wind to minimize land footprint.
- The project objective is to integrate co-location into energy models to study such scenarios.

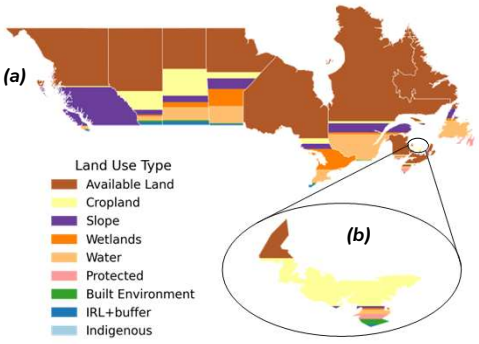


Figure 1. *Land Cover Shares per Province. (a) Canada. (b) Prince Edward Island [5-7].*

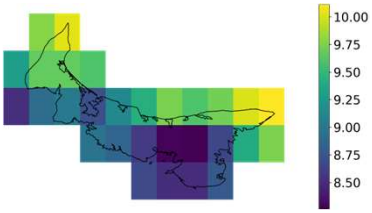


Figure 2. *Wind Speed for PEI (m/s) [8].*

Methods

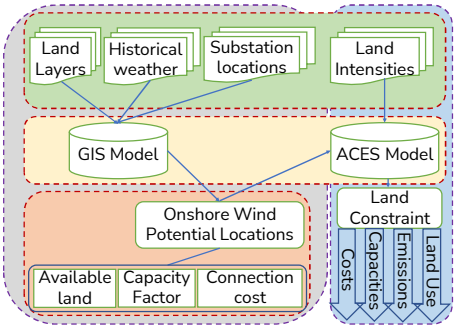


Figure 3. *Modeling Workflow.*

Land Limit (km2)	Base	Co-location
No limit	✓	✓
100 km^2	✓	✓
50 km^2	✓	✓
5 km^2	✓	✓
1 km^2	✓	✓

Figure 4. *Scenario Matrix.*

Results

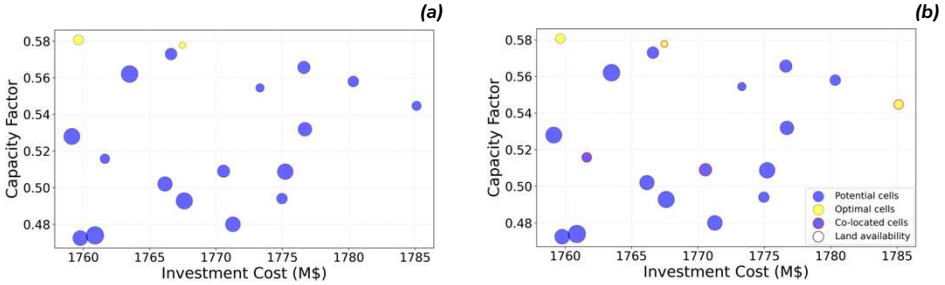


Figure 5. *Potential Onshore Wind Cells- (a) Base Scenario. (b) Co-location.*

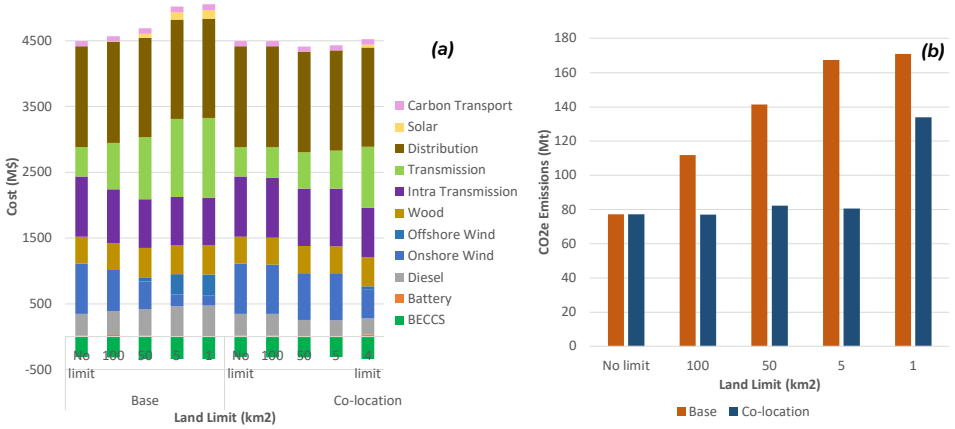


Figure 6. *Base vs. Co-location Scenario Results for (a) Electricity System Cost. (b) CO_2e Emissions. per Technology by 2050.*

Key Findings

- ✓ Combining energy and spatial models offers valuable features to study spatial scenarios such as co-location.
- ✓ Energy models without integrated co-location mislead future planning by favoring less land intensive technologies.
- ✓ Onshore wind and agriculture co-location in PEI is one of the most promising co-location opportunities in Canada.
- ✓ Lower costs and emissions are achievable under co-located and land-constrained scenarios.

References

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